In [1]:

```python
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate population
population_mean = 50
population_std = 10
population_size = 100000

population = np.random.normal(
    population_mean, population_std, population_size
)

# Step 2: Random sampling
sample_sizes = [30, 50, 100]
num_samples = 1000

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Step 3: Plot sampling distributions
plt.figure(figsize=(12, 8))

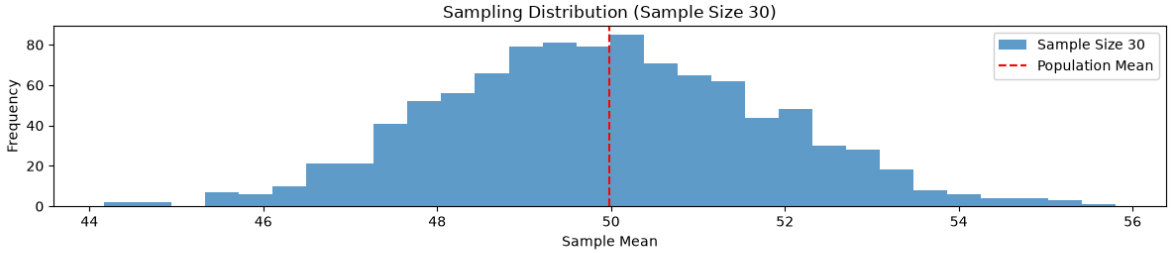
for i, size in enumerate(sample_sizes):
    plt.subplot(3, 1, i + 1)

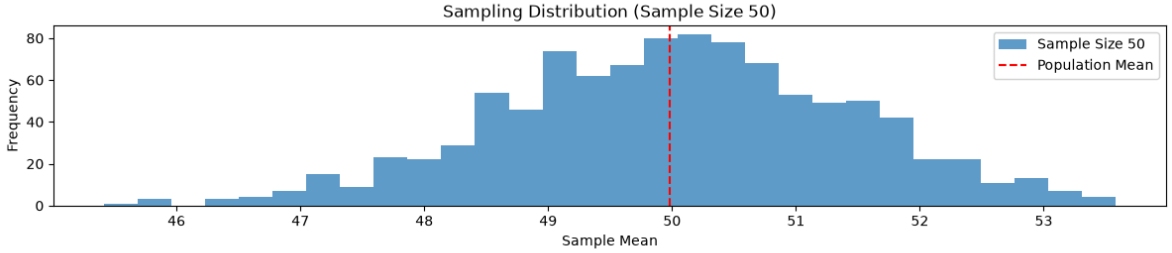
    plt.hist(
        sample_means[size],
        bins=30,
        alpha=0.7,
        label=f'Sample Size {size}'
    )

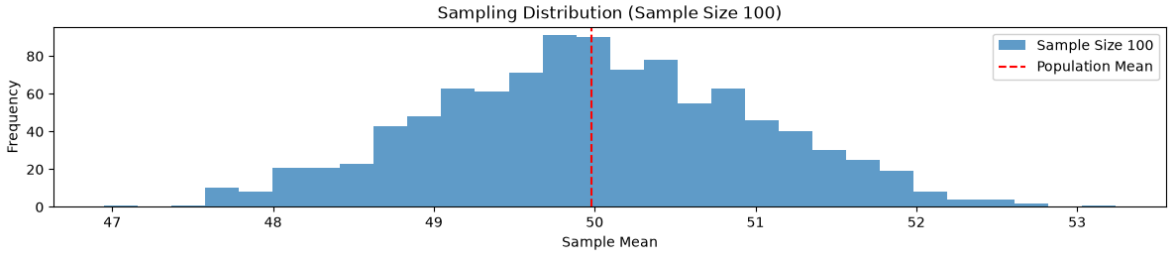
    plt.axvline(
        np.mean(population),
        color='red',
        linestyle='dashed',
        linewidth=1.5,
        label='Population Mean'
    )

    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```


Sampling Distribution (Sample Size 30)
Sample Size 30
Population Mean
Frequency
Sample Mean
0
20
40
60
80
44
46
48
50
52
54
56


Sampling Distribution (Sample Size 50)
Sample Size 50
Population Mean
Frequency
Sample Mean
0
20
40
60
80
46
47
48
49
50
51
52
53


Sampling Distribution (Sample Size 100)
Sample Size 100
Population Mean
Frequency
Sample Mean
0
20
40
60
80
47
48
49
50
51
52
53

In [ ]: